TDWG Standard · GBIF & OBIS · Open Biodiversity Data

Darwin Core (DwC), Data Publishing, Citation and IPT

Mobilizing, standardizing, and crediting global biodiversity records across the Open Science research lifecycle.

Begin Lecture →

Open Science Course

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The Problem: The Biodiversity Tower of Babel

Before standardization, natural history collections and field scientists recorded data in disparate silos:

🏷️ Taxonomy Chaos

Synonyms, typos, regional vernacular names, and conflicting authorities without persistent identifiers.

🗺️ Spatial Ambiguity

"Behind the old mill" vs degrees/minutes/seconds vs local UTM grids; lack of coordinate uncertainty metrics.

📅 Temporal Inconsistency

`04/05/06` (May 4th or April 5th?), arbitrary date intervals, time zones missing.

🎯 **The Consequence:** Global synthesis (climate change range shifts, extinction tracking, invasive species modeling) was virtually impossible without manual data wrangling.

What is Darwin Core (DwC)?

Darwin Core (DwC) is the international standard maintained by **TDWG (Biodiversity Information Standards)**.

- **A Glossary of Terms:** A stable vocabulary with precise URI definitions (e.g. `dwc:occurrenceID`, `dwc:scientificName`).
- **Pragmatic & Flexible:** Storage-independent (flat CSV, relational SQL, JSON-LD, RDF XML).
- **The Lingua Franca of Global Portals:** The foundational exchange format for **GBIF**, **OBIS**, and national biodiversity nodes.

🔗 **Essential Reference Guide:**
Consult the official Darwin Core Quick Reference Guide (dwc.tdwg.org) for every term definition, namespace, and normative examples.

Official TDWG Identity tdwg.org ↗

Standard:	**Biodiversity Information Standards (TDWG)**
Ratified:	October 2009 (Living standard)
Guide URL:	dwc.tdwg.org
Namespace:	http://rs.tdwg.org/dwc/terms/
Quick List:	dwc.tdwg.org/list/

"Darwin Core is primarily based on taxa, their occurrence in nature as documented by observations, specimens, samples, and related information."

GBIF Dataset Classes Overview

GBIF standardizes biodiversity publishing into **4 distinct dataset classes** (gbif.org/dataset-classes):

1. Occurrence

`Core: dwc:Occurrence`

Evidence of a species at a given time and place (museum specimens, citizen science observations, camera traps).

`Most common on GBIF`

2. Sampling-Event

`Core: dwc:Event + Occ Ext`

Standardized environmental monitoring, surveys, transects with quantifiable effort and non-detection (absence) data.

`Ecological time-series`

3. Checklist

`Core: dwc:Taxon`

Catalogues of taxa, national inventories, red lists, and taxonomic classifications without specific coordinate events.

`Taxonomic backbone`

4. Metadata-only

`Core: None (EML XML)`

Describing undigitized physical collections, historical literature, or projects awaiting electronic mobilization.

`Collection discovery`

GBIF Recommendations: Mandatory vs. Recommended Terms

Explore terms per dataset class based on gbif.org/dataset-classes and dwc.tdwg.org:

Occurrence | Sampling-Event | Checklist | Metadata-Only

All | ● Mandatory Only

Occurrence Datasets (Core: dwc:Occurrence)

GBIF Class Docs

Evidence of the presence, abundance, or absence of a biological organism at a specific location and date.

Darwin Core Term	GBIF Status	Definition & Purpose	Example Value
dwc:occurrenceID Occurrence	● Mandatory	Globally unique, persistent identifier for the occurrence record. ↳ Primary key; GBIF indexing fails without it	urn:uuid:8b3e8e2b-2e92-498c-851c-a1288c8eb25a
dwc:basisOfRecord Occurrence	● Mandatory	Controlled vocabulary denoting the specific nature of the record. ↳ Required controlled vocabulary for filtering	HumanObservation, PreservedSpecimen, MaterialSample

The Anatomy of a Darwin Core Record

A practical example of how a biological observation maps directly to standardized DwC terms:

```json
{
  "dwc:occurrenceID": "urn:uuid:8b3e8e2b-2e92-498c-851c-a1288c8eb25a",
  "dwc:basisOfRecord": "HumanObservation",
  "dwc:scientificName": "Bombus terrestris (Linnaeus, 1758)",
  "dwc:kingdom": "Animalia",
  "dwc:taxonRank": "species",
  "dwc:decimalLatitude": 50.0755,
  "dwc:decimalLongitude": 14.4378,
  "dwc:geodeticDatum": "WGS84",
  "dwc:coordinateUncertaintyInMeters": 15,
  "dwc:eventDate": "2024-06-12T10:15:00Z",
  "dwc:recordedBy": "Michal Torma",
  "dcterms:license": "http://creativecommons.org/publicdomain/zero/1.0/"
}
```

Mandatory Core: `occurrenceID`, `basisOfRecord`, `scientificName`, `eventDate`.

Strongly Rec: `decimalLatitude`, `decimalLongitude`, `geodeticDatum`, `coordinateUncertaintyInMeters`.

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